

301 TXOne Solutions

Basic Implementation Training Labs

Network Defense Solutions

Rev 3.5_251203

Global Training Services Team
TXOne Network Inc

3/15/2026

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Revision History

Version	Author	Date	Change
v2.1	Louis Liu	2023/02/15	1 st edition
v2.1b	Louis Liu	2023/03/05	2 nd edition, 2 attendees share 1 EdgeIPS
v2.1c	Louis Liu	2023/04/18	3 rd edition. Added Edge Lab 8: EdgeFire NAT settings
v2.2c	Louis Liu	2023/05/15	Using TXPI for MP installation
v2.2c	Chris Van Den Abbeele	2023/06/04	Minor cosmetic updates
v2.3	Louis Liu	2023/06/05	- EdgeOne Installation & Configuration - EdgeIPS 103 Installation & Configuration - Portable Inspector
v2.4	Louis Liu	2023/06/13	Split lab guide by solutions based on v2.3
v3.0	Louis Liu	2024/04/05	Win7 VM consolidation
v3.2	Chris Van Den Abbeele	2025/01/29	Revised for improved phrasing
v3.3	Louis Liu	2025/02/05	EdgeOne 2.4 , Edge device firmware 2.1
v3.4	Louis Liu	2025/06/26	Add Lab 8: Policy Rule Auto-Learning, in case there is no EdgeFire device available.
3.5	Louis Liu	2025/12/03	EdgeOne 2.4.18 upgrade to 2.5.18 Edge 102 G2 firmware 2.1 upgrade to 2.2

Product Versions

Solutions	Edition
EdgeOne	2.4.18 -> 2.5.18
EdgeIPS102	2.1.38 -> 2.2.11
EdgeFire	2.1.9

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1. Lab scenarios mapped to product features

Table 1 Lab scenarios mapped to product features

Solution	Lab scenario
EdgeOne	Lab1: EdgeOne Deployment 1. Installation, basic CLI commands
	Lab2: Initial setup / upgrade of EdgeOne and web UI walk thru 1. Administration functions: License, System Time, Updates, Account Management, Proxy 2. EdgeOne upgrade to the new version
EdgeIPS	Lab3: Edge Hardware, Installation and lab setup 1. DIN Rail Mounting 2. Power Connection 3. VMWare network configuration for lab exercises
	Lab4: EdgeIPS device initialization 1. Device settings 2. Operation mode 3. LED Status
	Lab5: Register EdgeIPS with EdgeOne for centralized management 1. Device System time setting 2. EdgeIPS sync to EdgeOne use passcode authentication method
	Lab6: Device Group Management 1. Device Groups 2. Group Settings
	Lab7: Policy Enforcement Configuration 1. Policy Enforcement operation modes 2. Basic Filter 3. Advanced Filter
	Lab8: EdgeIPS Policy Rule Auto-Learning 1. Rule Auto-Learning 2. Enforce policies based on learned results to block unexpected behavior

Solution	Lab scenario
EdgeFire	Lab9: EdgeFire initialization of settings and synchronization with EdgeOne 1. EdgeFire firmware upgrade 2. EdgeFire synchronization with EdgeOne using passcode authentication
	Lab10: EdgeFire NAT Settings 1. Many to One NAT 2. One to One NAT 3. Port Forwarding

2. Lab Environment content list

2.1 Virtual Machines

Table 2 Virtual Machines

Host	Server Name ZIP Filename	VMware Image Name	Hardware Specification	Purpose
A	LabConsole_xx xxx.zip	LabConsole_x xxx	CPU: 2 Core RAM: 4GB Hard Disk: 100G Network: NAT / Bridged (e.g. to VMnet11)	Main Console for labs
A	EdgeOne	edgeone_x.x. x.ova	CPU: 8 Core RAM: 16GB Hard Disk: 25G/80G Network: NAT / Bridged (e.g. to VMnet11)	EdgeOne server
B	Win7_xxxx.zip	Win7_xxxx	CPU: 2 Core RAM: 2GB Hard Disk: 60G Network: NAT / Bridged (e.g. to VMnet11)	Target for EdgeIPS, EdgeFire

2.2 Hardware/Software Requirements

Table 3 Hardware / Software Requirements

No	Items	Specification			
1	Laptop	OS	Windows 10 / 11		
		Hard Drive	65GB minimum		
		Memory	16GB Recommend		
		Network Adapter	WIFI NIC for Internet Access LAN NIC for Lab practice		
		Hard drive space requirements			
		Solutions	Space Requirements	Memory Requirements	VMs
		EdgeOne EdgeIPS EdgeFire	Host A: 48GB Host B: 20GB	Host A: 4.5GB Host B: 1GB	Host A: -LabConsole -EdgeOne Host B: -Win7
2	VMWare Workstation Pro 17	Hypervisor Minimum Hardware level = 16			
3	EdgeIPS	Model IPS-102-BP-G2-R-TX (2 attendees share 1 EdgeIPS device)			
4	EdgeFire	Model IEF-1012-TM (2 attendees share 1 EdgeFire device)			
5	Network Cable	2 UTP Network cables			

2.3 Lab network architecture

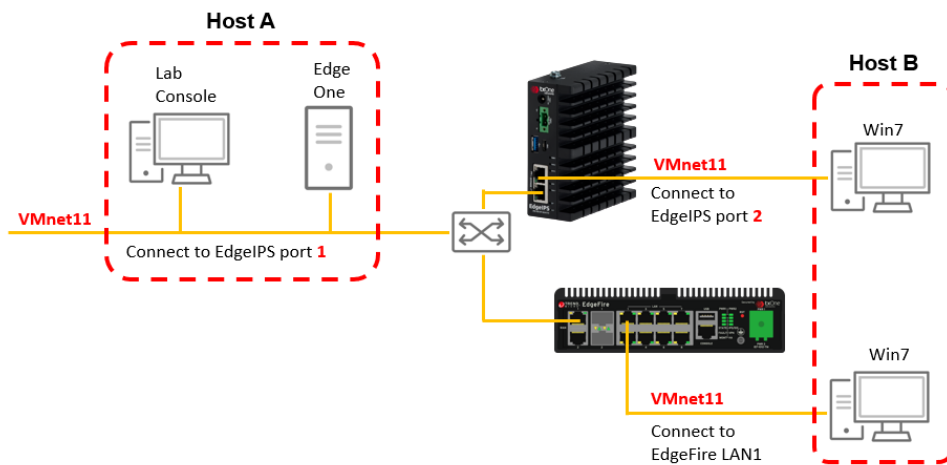


Figure 1 Lab network architecture

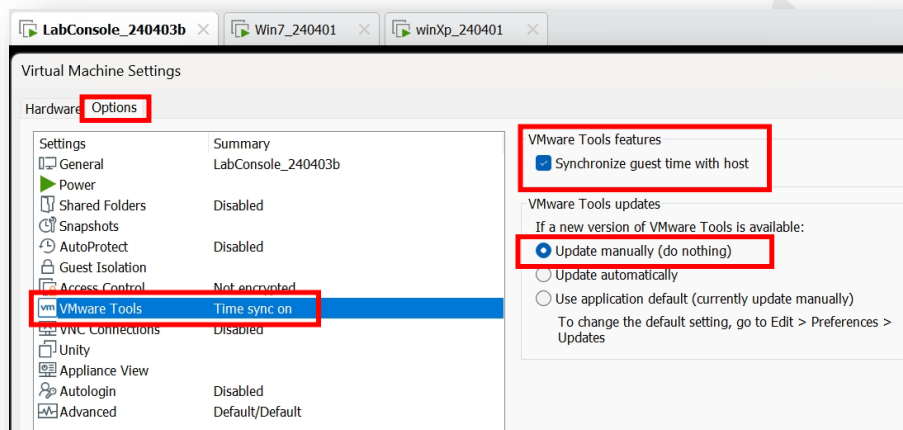
2.4 Credentials

Table 4 Lab Credentials

No	VM Image	IP	Account / Password
1	LabConsole	192.168.23.196	txone/12345678
2	EdgeOne vx.x	192.168.23.21	Web: txone/P@ssw0rd Server: admin/P@ssw0rd
3	Win7	EdgeIPS: 192.168.23.24 EdgeFire: 192.168.100.25	txone/12345678
4	EdgeIPS	192.168.23.200	txone/P@ssw0rd
5	EdgeFire	192.168.23.26	txone/P@ssw0rd

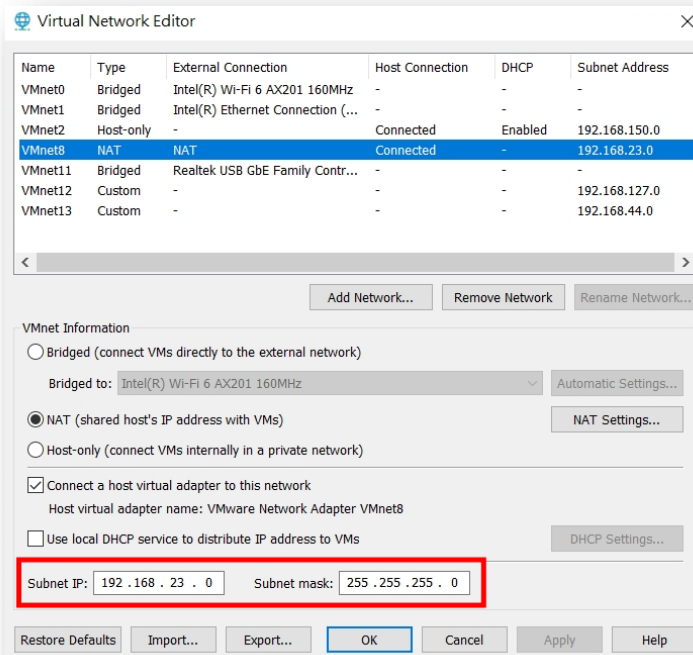
2.5 VMWare VMTools update Settings

- Edit the Settings of each VM
- Choose the **Options** tab
- Select the **VMWare Tools** option
- Check **“Synchronize guest time with host”**
- Under VM Tools updates, select the option **“Update manually (do nothing)”**

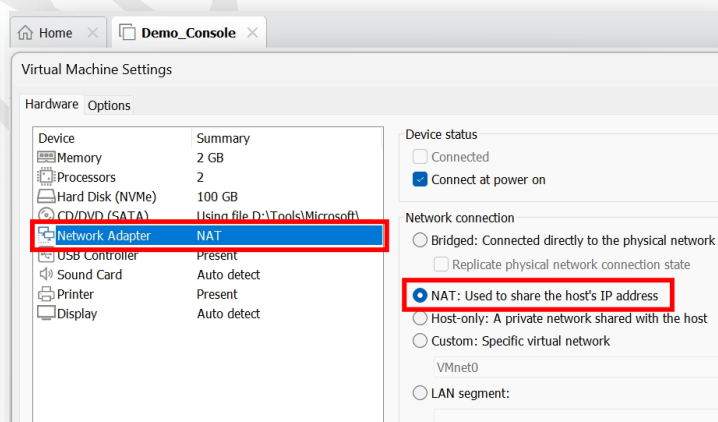


2.6 VMWare Networks Settings

- Make sure your laptop can access the Internet via its wireless interface. You will also need a UTP network interface, which will be used to connect to the EdgeIPS.
- In VMware workstation, open the Virtual Network Editor and configure the VMWare NAT network to 192.168.23.0 /24.



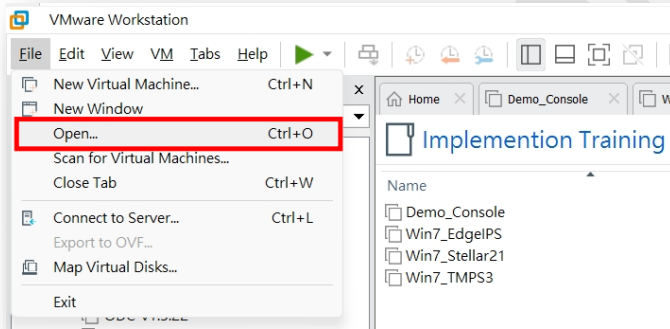
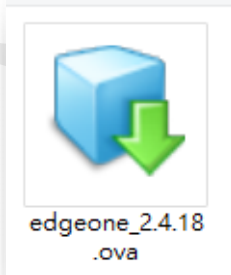
- Set the Network Adapter to NAT for LabConsole.
 - The default NAT gateway setting is **192.168.23.2**.
 - LabConsole and EdgeOne use **8.8.8.8** as DNS server.

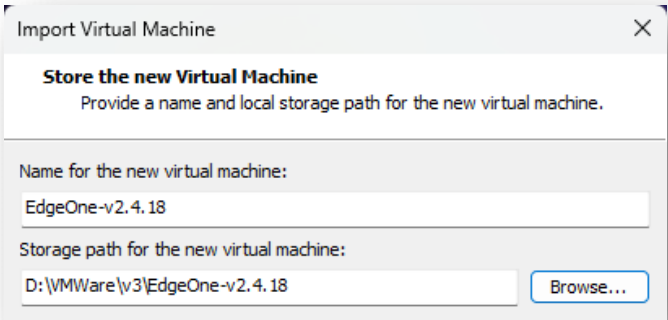
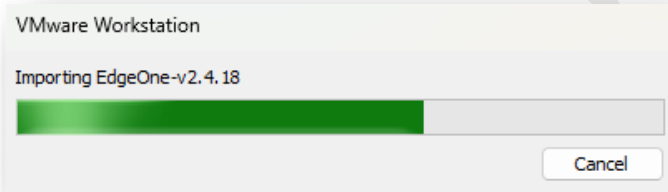
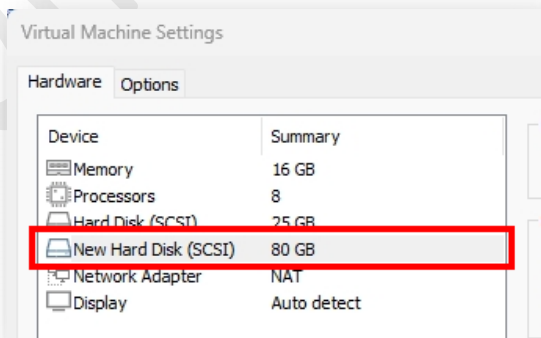


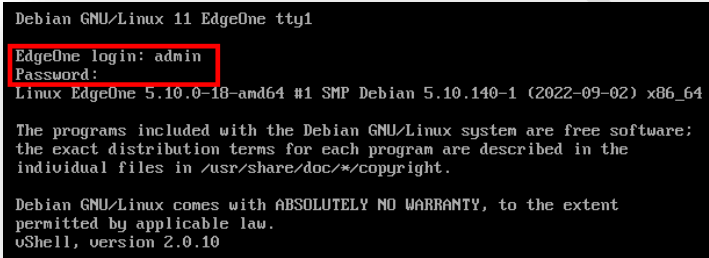
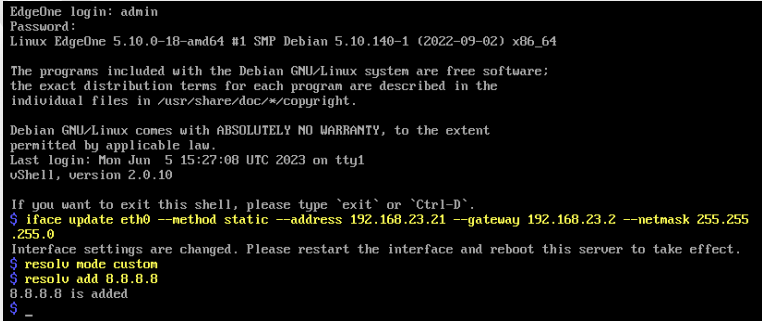
3. EdgeIPS / EdgeFire Lab Guide

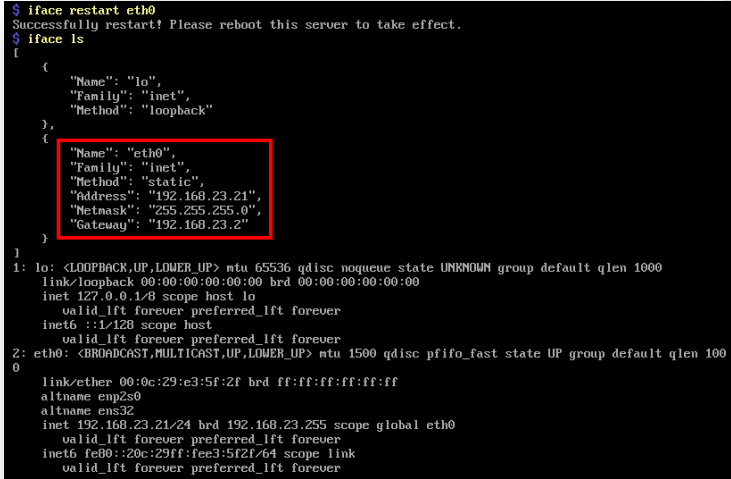
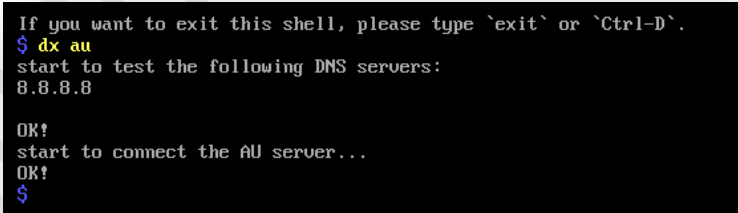
Lab1: EdgeOne Deployment

Table 5 EdgeOne Deployment

Items	Descriptions
Application target	Host A
Steps	Import the EdgeOne OVA VMWare Image - Launch VMWare Workstation and open the EdgeOne OVA image.  - Select the OVA image file "edgeone_xxx.ova" from Training package.  - Name the Virtual Machine "EdgeOne-vxxx" and select Storage Path for the new virtual machine. Note: If you are using OneDrive on your host, ensure that the storage path of the VM is not a directory synchronized with OneDrive. Every time the OneDrive agent syncs with the server, it can disrupt the integrity of the virtual machine, leading to numerous errors in the EdgeOne CLI.

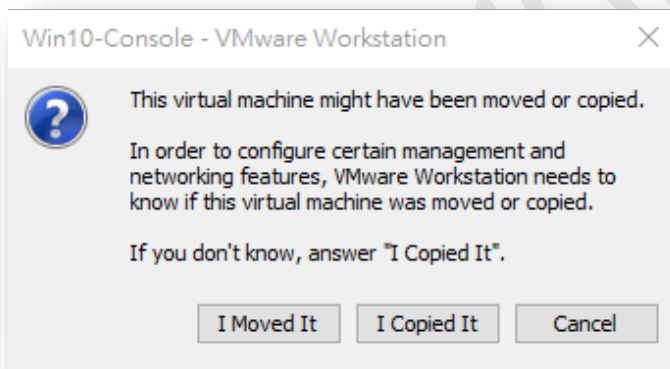
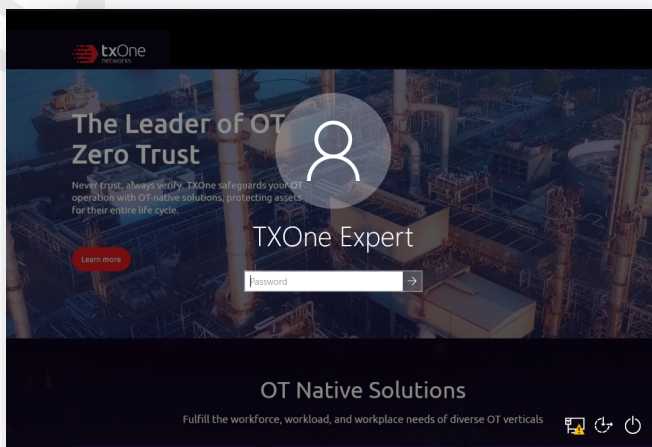
Items	Descriptions
	 4. The EdgeOne OVA image import process will take a moment.  5. After the OVA image import has completed. Add the 2nd hard drive to the EdgeOne virtual machine. Allocate at least 80GB for the 2nd hard drive (SCSI Disk type). Do not worry if you do not have that space on your laptop. This hard disk is “thin provisioned”. This means it will only consume the space that it needs, but it needs to be configured as 80GB 

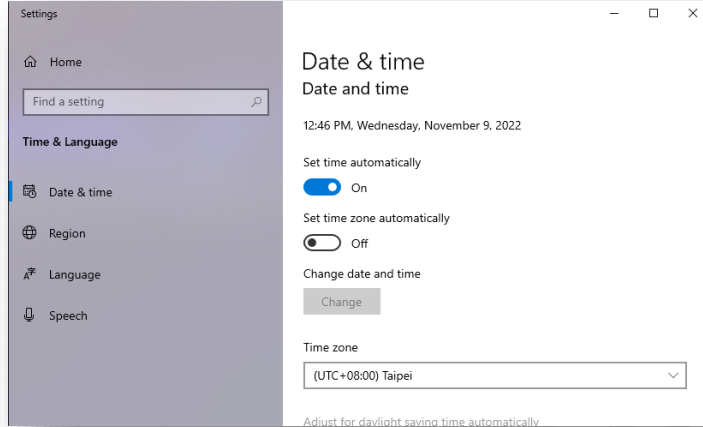
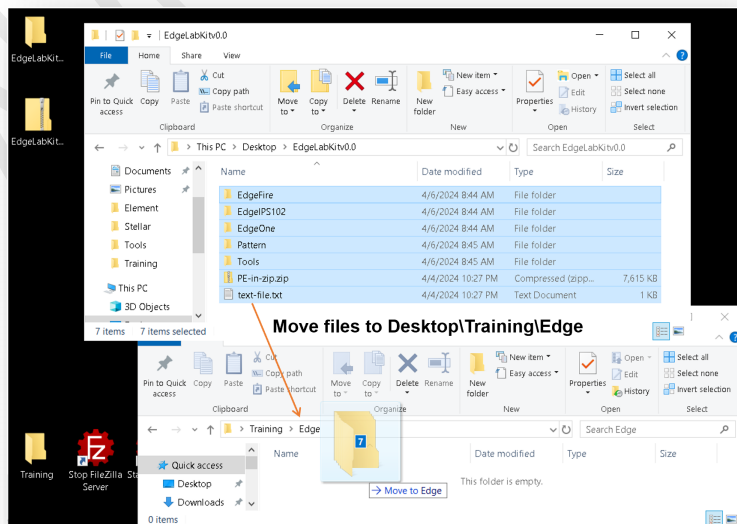
Items	Descriptions
	EdgeOne Server Network configuration - Make sure that the VMWare Network Adapter is set to NAT and turn on the EdgeOne virtual machine. Log in to the EdgeOne web UI with the following credentials. User: admin Password: txone  - Run the “oobe” command to change the root password to “P@ssw0rd”. - Use the new credentials “admin/P@ssw0rd” to Log in to the EdgeOne server again. - Run the following commands in the restricted oobe shell to configure the network interface: iface update eth0 --method static --address 192.168.23.21 --gateway 192.168.23.2 --netmask 255.255.255.0 - Setup DNS resolv mode custom resolv add 8.8.8.8 

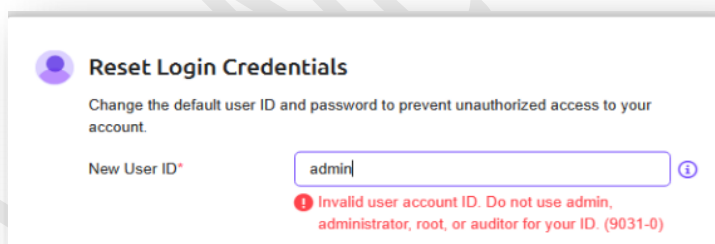
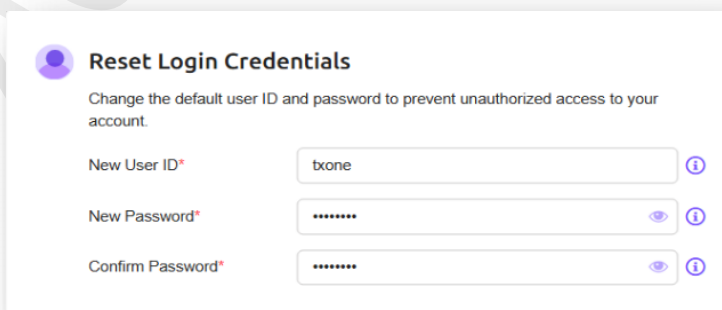
Items	Descriptions
	6. Check the network settings and restart the EdgeOne virtual machine: <pre>iface restart eth0</pre> iface ls reboot 7. Note: if it takes a long time to execute the “iface restart eth0” command, then please check your IP address settings.  <pre>\$ iface restart eth0 Successfully restart! Please reboot this server to take effect. \$ iface ls { { "Name": "lo", "Family": "inet", "Method": "loopback" }, { "Name": "eth0", "Family": "inet", "Method": "static", "Address": "192.168.23.21", "Netmask": "255.255.255.0", "Gateway": "192.168.23.2" } } 1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000 link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 inet 127.0.0.1/8 scope host lo valid_lft forever preferred_lft forever inet6 ::1/128 scope host valid_lft forever preferred_lft forever 2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000 link/ether 00:0c:29:c3:5f:2f brd ff:ff:ff:ff:ff:ff altname emp2s0 altname ens32 inet 192.168.23.21/24 brd 192.168.23.255 scope global eth0 valid_lft forever preferred_lft forever inet6 fe80::20c:29ff:fe3:5f2f/64 scope link valid_lft forever preferred_lft forever</pre> 8. Log in to EdgeOne and enter the “dx au” command to test the connection with update server. The command will show the result of the connection to the DNS server and the AU server.  <pre>If you want to exit this shell, please type `exit` or `Ctrl-D`. \$ dx au start to test the following DNS servers: 8.8.8.8 OK! start to connect the AU server... OK! \$</pre>
Expected Result	The EdgeOne server has been successfully imported and is up and running.

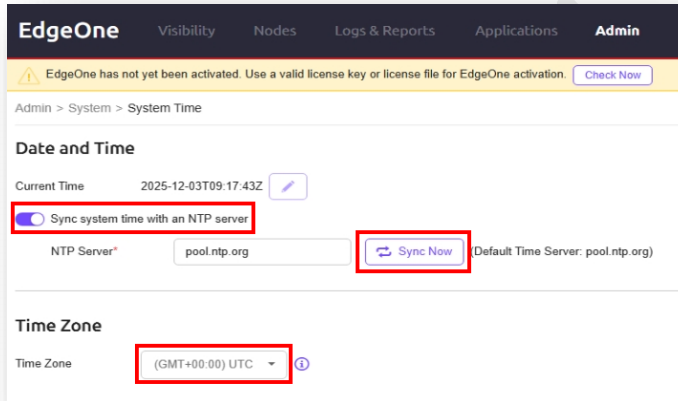
Lab2: Initial setup / upgrade of EdgeOne and web UI walk thru

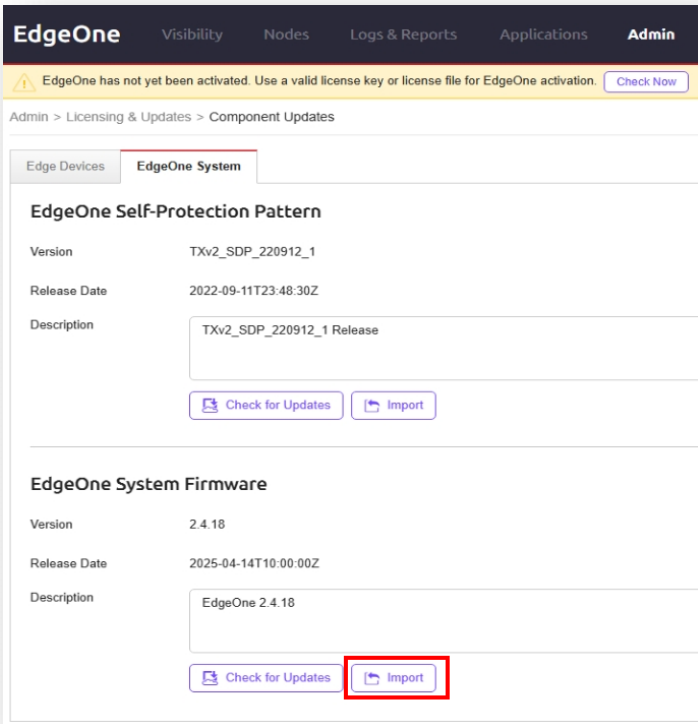
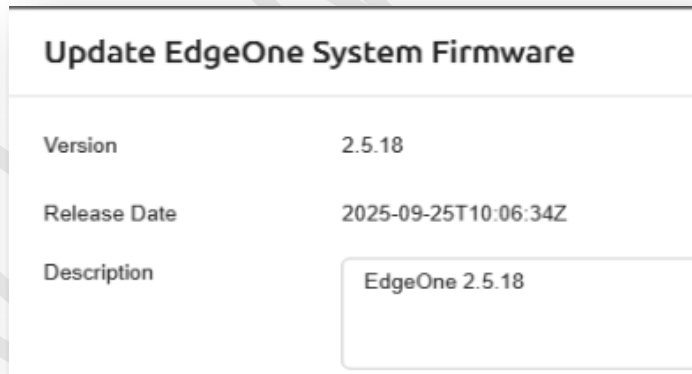
Table 6 Initial setup / upgrade of EdgeOne and web UI walk thru

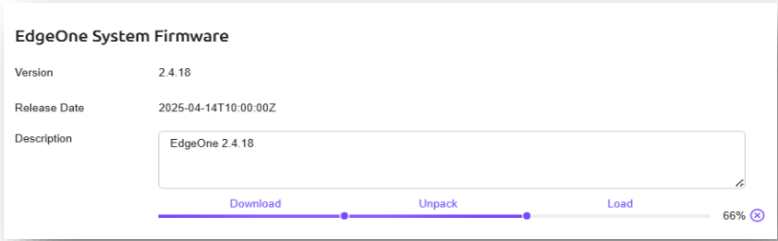
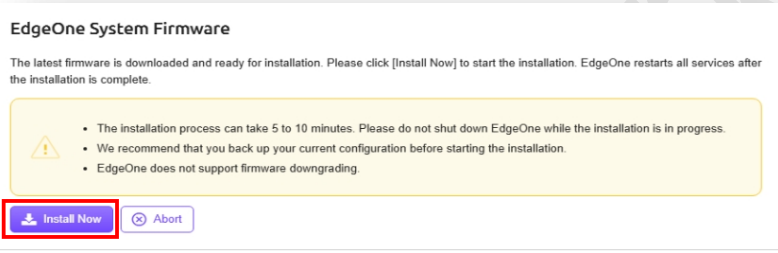
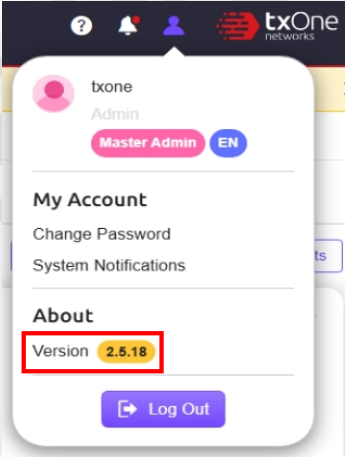
Items	Descriptions
Application target	Host A
Steps	Configure access to the EdgeOne web UI from the VM LabConsole 1. In VMWare Workstation, open the LabConsole VM. 2. Revert the VM LabConsole to the snapshot "ReadyforLabs". 3. Set the LabConsole VM network adapter to NAT, then power on the VM. 4. if prompted for the first boot, then click "I Moved It"  5. Log in using the following credentials. 6. User: txone (TXOne Expert) 7. Password: 12345678 

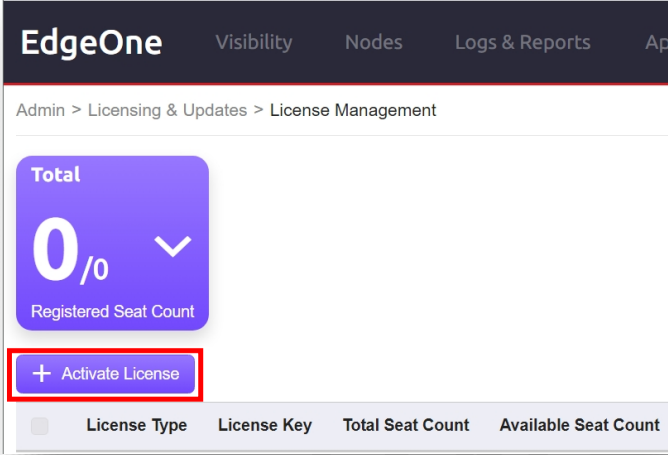
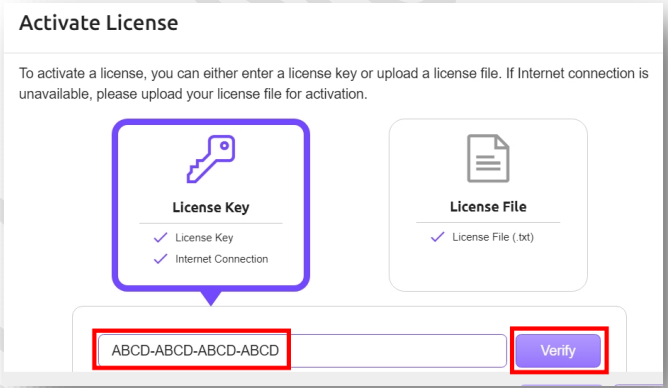
Items	Descriptions
	8. Configure the Windows OS Time Zone of the LabConsole VM 9. For this training, set the time zone to UTC  10. Copy the “EdgeLabKitVx.y.zip” (v2.5 or above) that was shared with you to the desktop of the LabConsole and expand it (right click the file and “Extract All”). It will create the “EdgeLabKitVx.y.zip” folder on the desktop. 11. Move all subdirectories from that folder to the Edge folder, which is in the Training Folder on the Desktop, as shown in the screenshot below. 12. The training folder is hosted on a webserver running in this VM. We will access it from other VMs to download the required files. 13. Move the files “PE-in.zip” and “text-file.txt” from \Desktop\Training\Edge to \Desktop\Training 

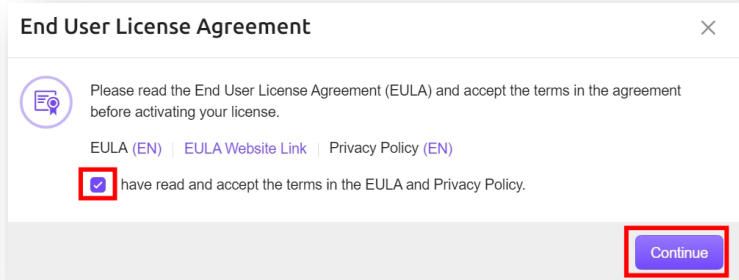
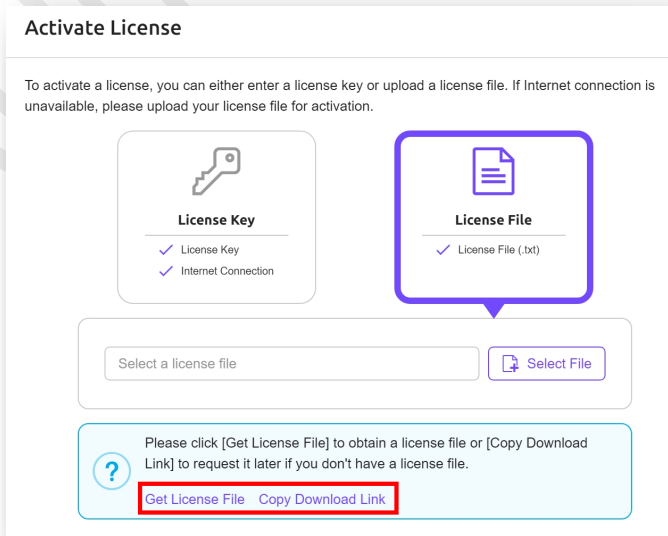
Items	Descriptions
	14. Launch the Chrome browser from the Windows taskbar and navigate to: https://192.168.23.21. 15. Login in using admin / txone 16. Click Confirm to accept the EULA and OT Intelligent Trust.  17. You will be prompted to create a new admin user and set password. 18. The following usernames are not allowed: “admin, administrator, root and auditor”  19. Create the user txone with the password P@ssw0rd, then log in again. 

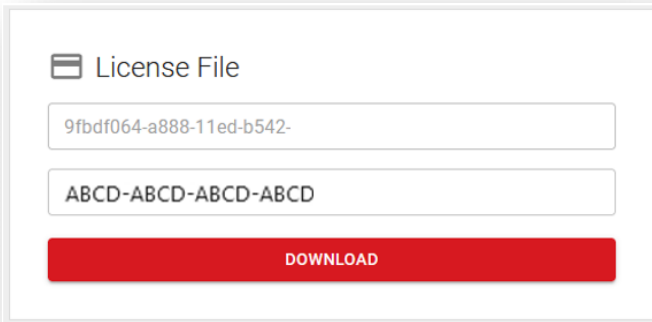
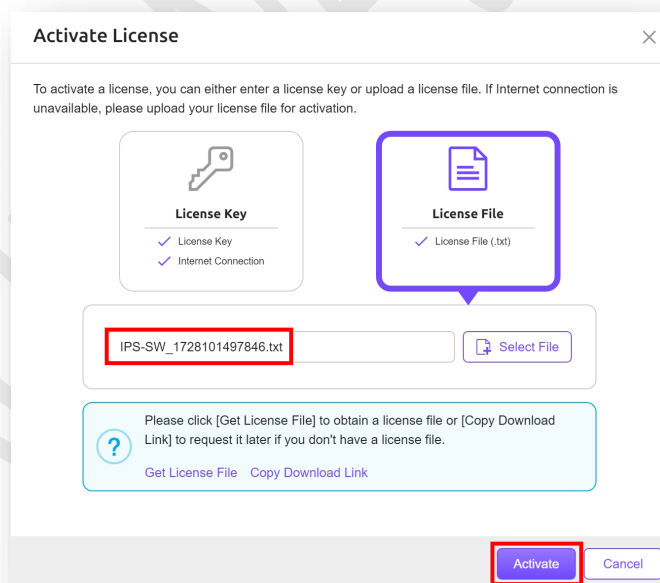
Items	Descriptions
	System Time setting - Go to Admin > System > System Time - Set date / time manually or synchronize with an NTP server. - For these labs, leaving the time zone at UTC is acceptable. Note: For real deployments, it is recommend configure the local time zone.  Upgrade EdgeOne - Go to Admin > Licensing & Updates > Component Update - Select the EdgeOne System tab and click Import under the EdgeOne System Firmware session. - Choose the latest version of EdgeOne update firmware file from the folder Desktop\Training\Edge\EdgeOne. - Upgrade from 2.0 ~ 2.3 -> 2.4 and above: use egone.fw_xxx.acf - Upgrade from 2.4+ ~ 2.4 and above: use egone.cms.v2_xxx.acf

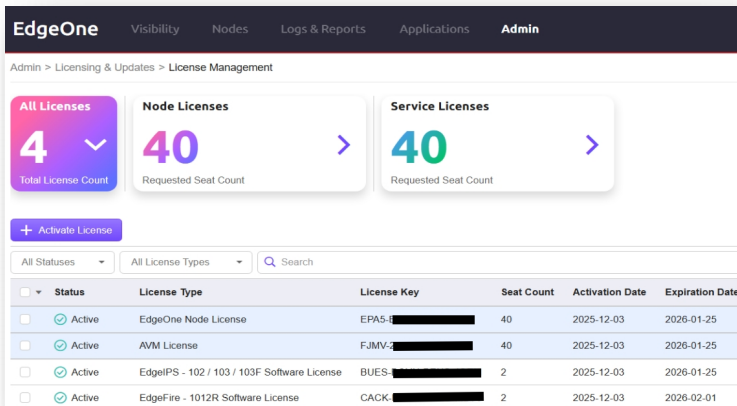
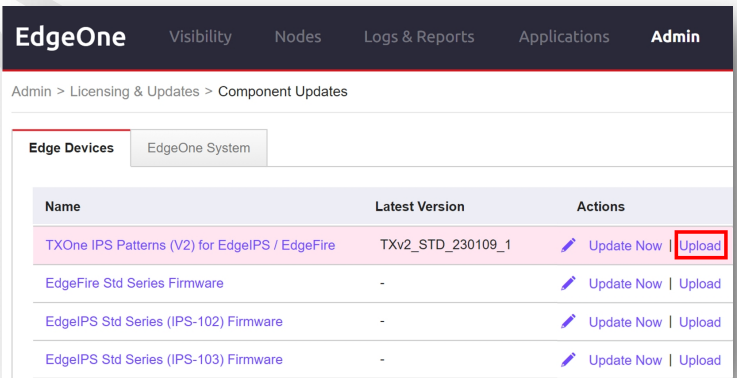
Items	Descriptions
	 4. EdgeOne will validate the firmware integrity. 

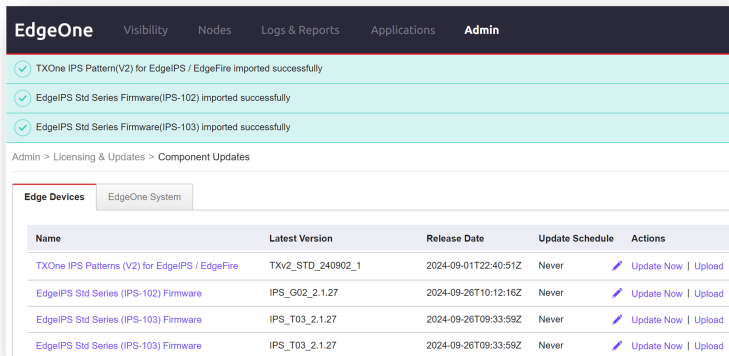
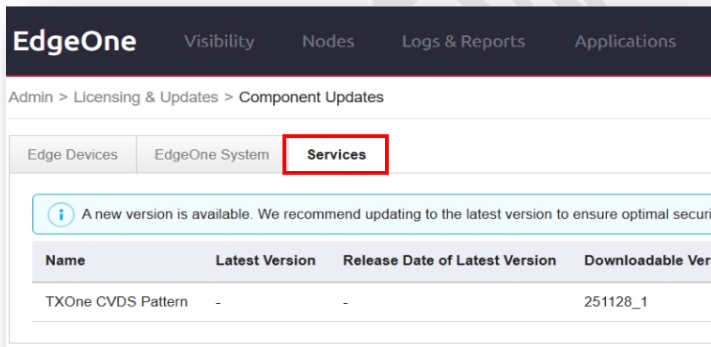
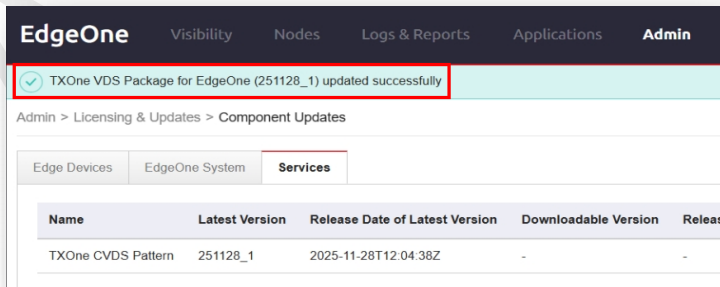
Items	Descriptions
	5. EdgeOne displays progress stages: Download, Unpack and Save.  6. Click Install Now. Installation typically completes in under 1 minute.  7. After reboot, go to Log in User > About to verify the version. 

Items	Descriptions
	Activate a license using the “online” method. - In EdgeOne go to Admin > Licensing & Updates > License Management and click +Activate License.  - Click Online license activation. Enter the EdgeOne Node license key that will be provided by your trainer and click Verify. 

Items	Descriptions
	3. Accept the End User License Agreement  The dialog box titled "End User License Agreement" contains a message: "Please read the End User License Agreement (EULA) and accept the terms in the agreement before activating your license." Below this, there are links for "EULA (EN)", "EULA Website Link", and "Privacy Policy (EN)". A checkbox is checked, and the text "have read and accept the terms in the EULA and Privacy Policy." is displayed. A "Continue" button is at the bottom right. Activate a license using the Offline Activation (License File method): 4. For air-gapped EdgeOnes, Offline license activation can be used. Let's use this option to add the licenses for EdgeIPS, EdgeFire and AVM (Asset Vulnerability Management). 5. Choose License File then click Get License File or Copy Download link depending on your Internet connection. <u>Get License File:</u> Choose this option if EdgeOne web UI has Internet access. <u>Copy Download Link:</u> If EdgeOne web UI has no Internet access, copy this link to a client that can access the internet.  The "Activate License" dialog box provides instructions: "To activate a license, you can either enter a license key or upload a license file. If Internet connection is unavailable, please upload your license file for activation." It shows two options: "License Key" (with a key icon and checkmarks for "License Key" and "Internet Connection") and "License File" (with a document icon and a checkmark for "License File (.txt)"). Below these is a "Select a license file" input field and a "Select File" button. At the bottom, a message says: "Please click [Get License File] to obtain a license file or [Copy Download Link] to request it later if you don't have a license file." The "Get License File" and "Copy Download Link" buttons are highlighted with a red box.

Items	Descriptions
	6. EdgeOne in our lab environment is connected via NAT to your host and thus has internet access. So, click Get License File. Enter an EdgeIPS software license key and download the license file. In an air-gapped environment, you would copy the download link to a computer with internet access, to get the license file from there.  7. Return to EdgeOne web UI, select License File and click Apply to upload the license file to EdgeOne.  8. Repeat steps 5 and 6 to get the EdgeFire Software license and AVM license.

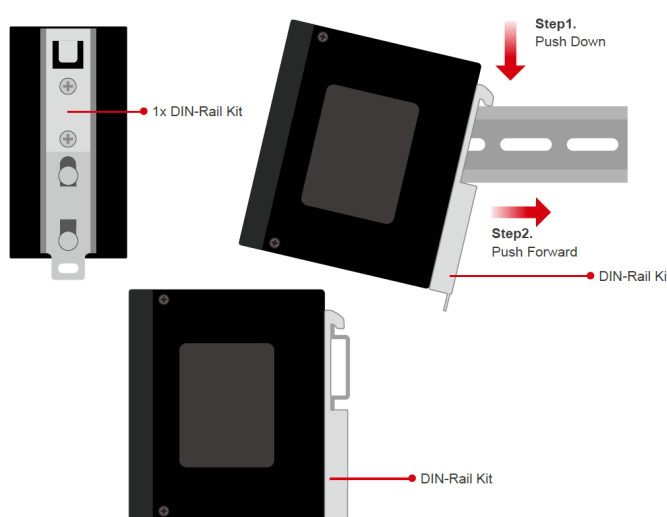
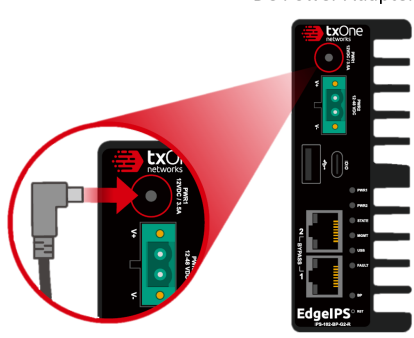
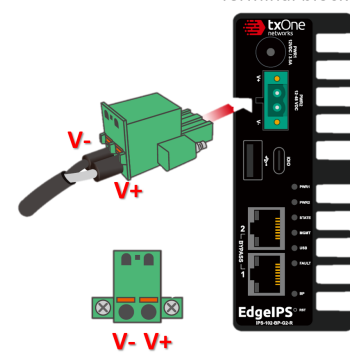
Items	Descriptions
	9. Verify the license list. There should be an EdgeOne Node License, EdgeIPS Software License and EdgeFire Software License. 10. You will notice that when you activate the EdgeOne license, any licenses that are already bound to that EdgeOne will be activated automatically if they were previously activated by someone else.  Update EdgeIPS firmware and patterns 11. Go to Admin > Licensing & Updates > Component Updates 12. Select the Edge Devices tab, find the Action column and: - Click Update Now for “TXOne IPS Pattern for EdgeIPS / EdgeFire” - Click Upload for “EdgeIPS Std Series Firmware for EdgeIPS-102” and upload the file from Desktop\Training\Edge\EdgeIPS102 - Click Upload for EdgeFire Firmware and upload the file from Desktop\Training\Edge\EdgeFire 

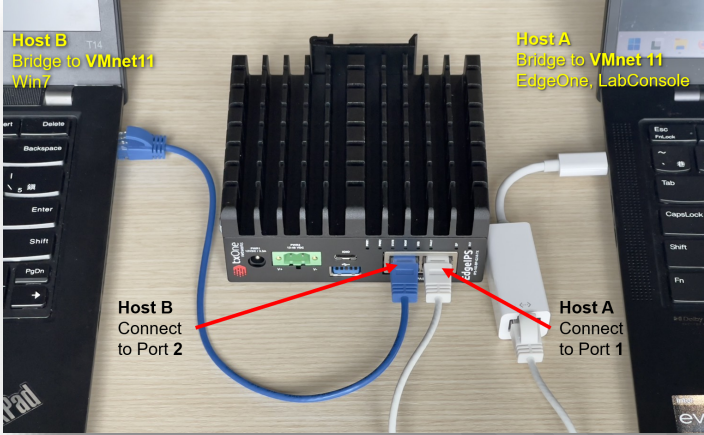
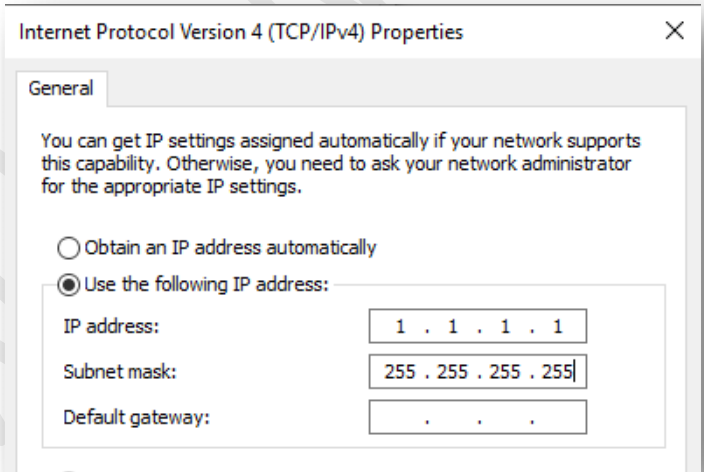
Items	Descriptions
	13. Notice the green banners indicating when the software has been successfully imported. The Device tab will display the latest versions and release dates.  14. Select the Services tab, then find the Actions column.  15. Click Update Now to download the latest version of CVDS pattern from TXOne Active Update server. 

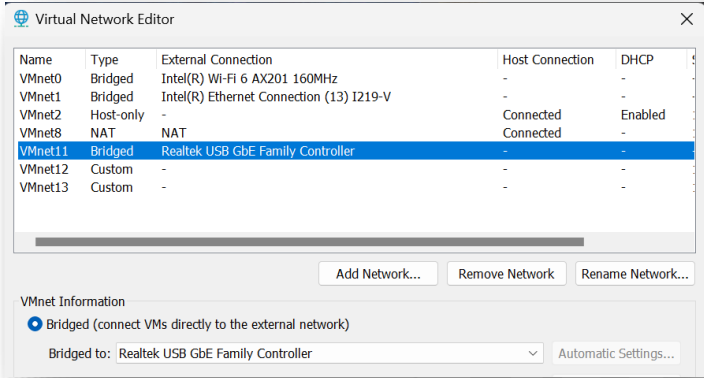
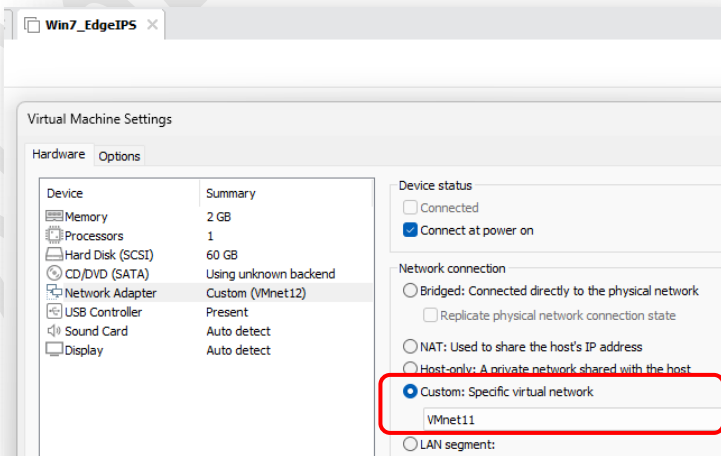
Items	Descriptions
	16. Now change the VMware network adapter for the EdgeOne server and the LabConsole to bridged (VMnet11).
Expected Result	EdgeOne configuration tasks completed. - Date / Time settings and Time Zone setup. - EdgeOne upgraded to latest version - Licenses activated. - Device components updated.

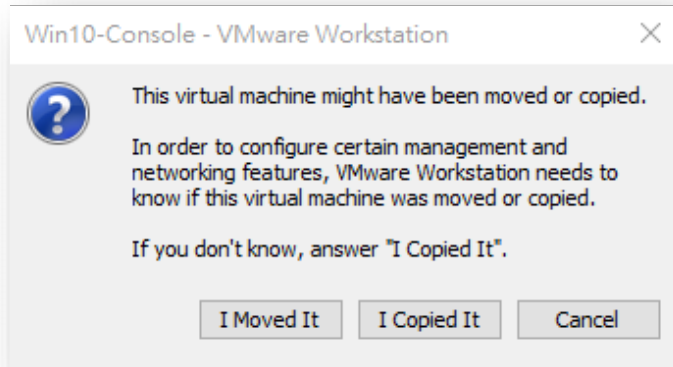
Lab3: EdgeIPS Hardware, Installation and Lab setup.

Table 7 EdgeIPS Hardware, Installation and Lab setup

Items	Descriptions
Application target	Host A and B
Steps	Din Rail Mounting (Informational)  Power Supply Connection - EdgeIPS 102 G2 supports dual power input. - Verify the power supply polarity (V+, V-) before connecting the power socket to EdgeIPS. DC Power Adapter  Terminal block 

Items	Descriptions
	EdgeIPS Lab network wiring and VMWare configuration Review the EdgeIPS hardware network wiring diagram. Do not turn on the EdgeIPS yet.  (Optionally) Assign physical host IP address 1.1.1.1/32 for Host A LAN Adapter and physical host IP address 1.1.1.2/32 for Host B LAN Adapter. This provides additional network isolation with the Lab network.  Bridge the VMWare network adapter to the physical network interface. In VMware workstation, go to the VMWare Virtual Network Editor. Bridge VMnet11 to your physical UTP LAN Adapter. This is an important step. Make sure to bridge to the right physical network adapter. (no virtual adapter, no wifi adapter,...)

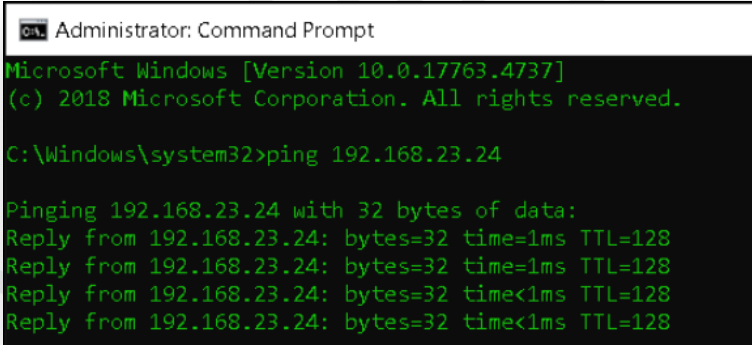
Items	Descriptions
	Note: Bridging a VMnet (virtual) network adapter to a specific physical LAN adapter may not be possible if one of the VMnet adapters is set to “Auto” mode. To resolve this, change the adapter from “Auto” to “Bridged” and assign it to the specific physical network card connected to your physical UTP port.  5. Now that the Virtual Network has been configured, connect your VMs to the right virtual network. 6. Host A bridged (VMnet11): EdgeOne, LabConsole (Stop all other VMs on host A) 7. Host B bridged (VMnet11): Win7 (Stop all other VMs on host B) 

Items	Descriptions
	8. In VMWare, open the VM images Win7 (do this on Host B!) 9. Restore the VM Win7 to snapshot ReadyforLabs. In VMware workstation, double-check the Network adapter of this VM: it must be set to bridged (VMnet11). 10. Power on the VM. Click "I Moved It" when this is the first power on of this VM  11. Log in using the following credentials. User: txone (TXOne Expert) Password: 12345678
Expected Result	EdgeIPS Wiring completed and VMWare network environment configuration completed.

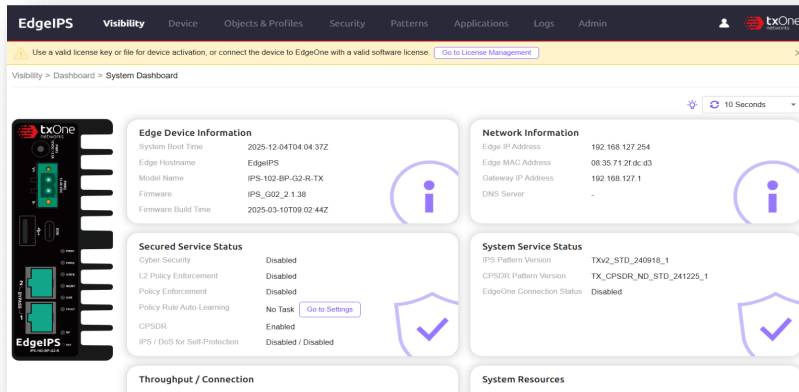
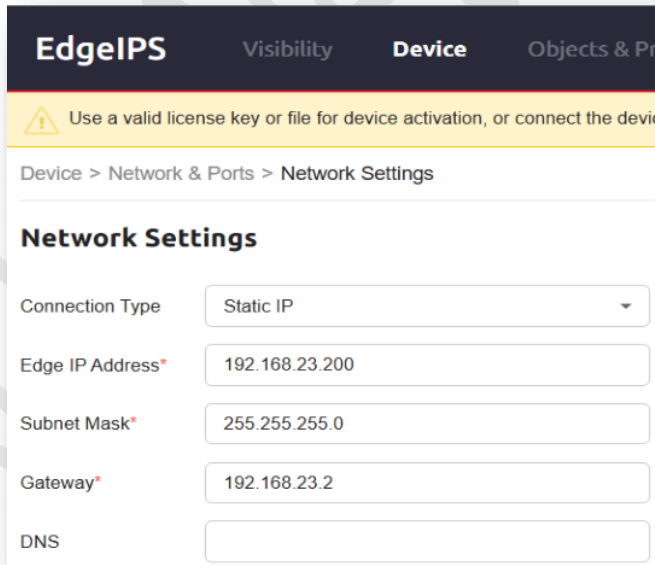
Lab4: EdgeIPS device initialization

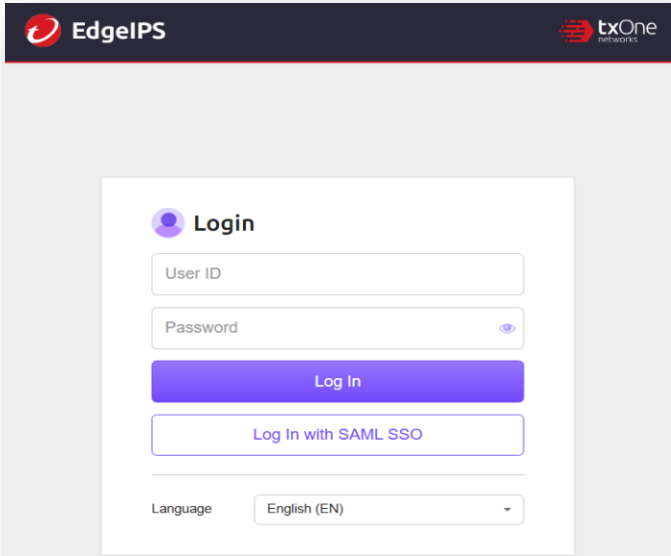
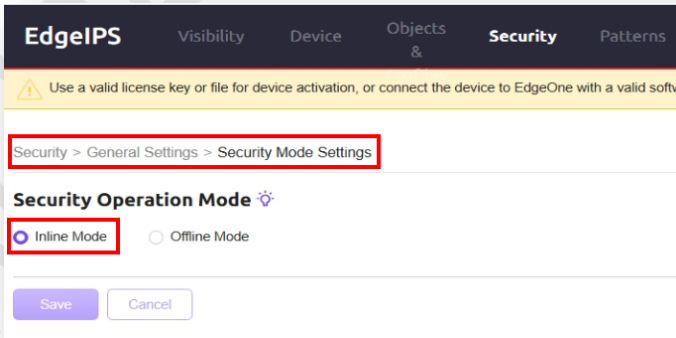
Table 8 EdgeIPS device initialization

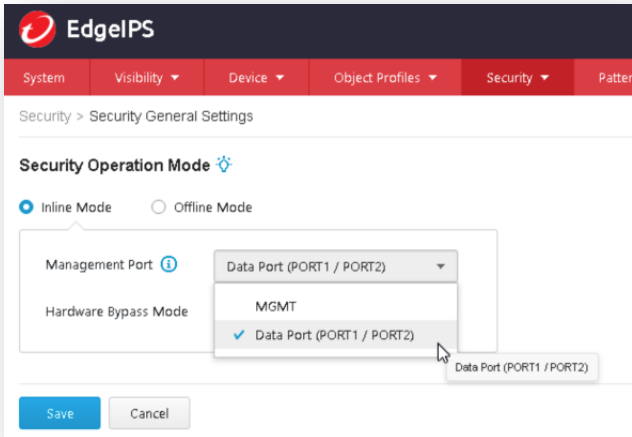
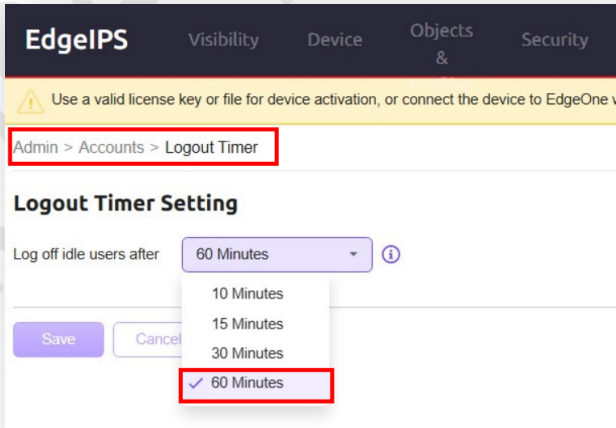
Items	Descriptions
Application target	Host A
Steps	- Turn on the EdgeIPS (Connect the power supply plug to the EdgeIPS). During power-on, the “PWR1” and “BYPASS” LEDs will be on. The EdgeIPS is still booting and it is connecting Port1 to Port2 without any traffic inspection. A photograph of the EdgeIPS device (model IPS-102-BP-G2-R-TX) from a top-down perspective. A red power cable is plugged into the PWR1 port (12VDC / 5A). The PWR1 and BYPASS LEDs are illuminated. The device has a black casing with various ports and labels. - Only the “PWR1” LED should be on. A photograph of the EdgeIPS device (model IPS-102-BP-G2-R-TX) from a top-down perspective. The PWR1 LED is illuminated, while the BYPASS LED is not. The device has a black casing with various ports and labels. - Factory-reset the EdgeIPS Before using the EdgeIPS, make sure to reset it (it may have been used by another student in another training). It may have been set to an IP that we don't know, and it may have policies that we don't want. A photograph of the EdgeIPS device (model IPS-102-BP-G2-R) from a top-down perspective. A red circle highlights the RST button, which is located next to the FAULT and BP LEDs. The device has a black casing with various ports and labels.

Items	Descriptions
	4. To factory reset the EdgeIPS, follow these steps: - Insert a paperclip into the reset hole and press the switch. - Keep pressing for a few seconds until the "STATUS" LED starts blinking slowly. Continue holding until it blinks faster. At this point, the device has reset itself, and you can release the button. - Avoid using a thick paperclip, sharp objects, or excessive force to prevent damage to the reset switch. 4. Log in to the VM LabConsole (on Host A), open cmd.exe and ping 192.168.23.24 (Win7). Make sure you are on Host A and the Win7 is only running on Host B (It must not run on Host A as well.) This ping will test the network connection between the LabConsole on Host A and Win7 on Host B.  5. If you have an EdgeIPS 103, which has a dedicated Management port, now connect the cable from Host A to the Management Port of the EdgeIPS103. We will change it later to the data port. 6. Launch the Chrome browser from LabConsole VM and navigate to: https://192.168.127.254. This should be reachable because the LabConsole VM has a secondary IP address on this network. You can check this by running "ipconfig".

Items	Descriptions
	7. Launch the Chrome browser from LabConsole VM and navigate to: https://192.168.127.254. This should be reachable because the LabConsole VM has a secondary IP address in this network. You can check this by running “ipconfig”. 8. Log in to the EdgeIPS device web console using following credential and accept the EULA. User: admin Password: txone 9. You will be prompted to create a new user and set password. Create the following user: txone with the password: P@ssw0rd 10. Re-Log in to the EdgeIPS device web UI using the new credentials “txone/P@ssw0rd”. 11. The EdgeIPS Device Web UI will show up when Log in was successful.

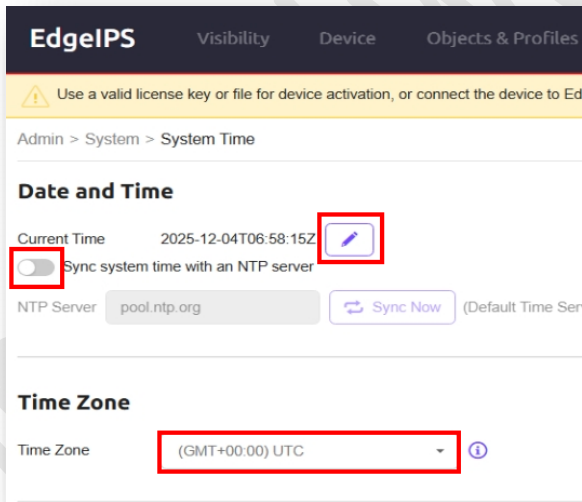
Items	Descriptions
	 12. Go to Device > Network & Ports > Network Settings to configure a new device IP address of the EdgeIPS. - Device IP address: 192.168.23.200 - Netmask: 255.255.255.0 - Gateway: 192.168.23.2  13. Click Save. 14. Re-log in to the EdgeIPS device web UI using new device IP address,

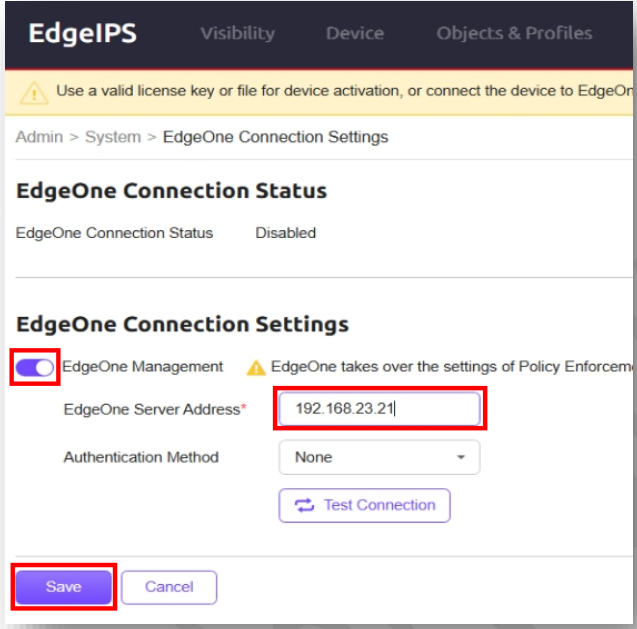
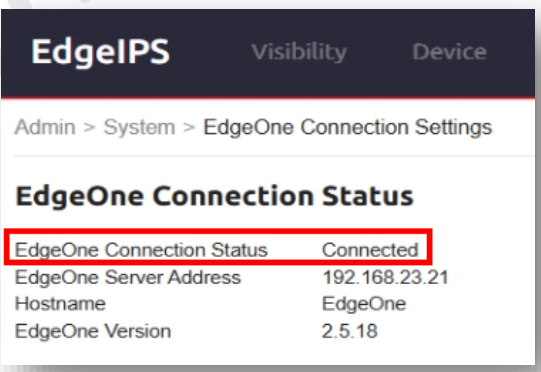
Items	Descriptions
	https://192.168.23.200 or simply click EdgeIPS bookmark.  The image shows the EdgeIPS login page. It has a dark header with the EdgeIPS logo and txOne networks logo. The main content area is white and contains a 'Login' section with fields for 'User ID' and 'Password', a 'Log In' button, and a 'Log In with SAML SSO' button. There is also a 'Language' dropdown menu set to 'English (EN)'. 15. Set Security Operation Mode to “Inline Mode” 16. Go to Security > Security General Settings 17. Security Operation Mode set to “Inline Mode” 18. (informational) If the EdgeIPS is used in offline mode, you need to assign the Management Port (port1 or port2)  The image shows the EdgeIPS Security Mode Settings page. The breadcrumb trail is 'Security > General Settings > Security Mode Settings'. The 'Security Operation Mode' section has two radio buttons: 'Inline Mode' (selected) and 'Offline Mode'. There are 'Save' and 'Cancel' buttons at the bottom. 19. If you are using an EdgeIPS103, also change the Management Port to one

Items	Descriptions
	of the Data ports (unless you have brought a switch with you and want to keep using the dedicated management port)  20. Click Save and move the network cable from the Management Port to Port1. You should now be able to reach the management interface of the EdgeIPS 103 through its Data Ports. 21. Set the Logout Timer to 60 minutes Navigate to Admin > Accounts > Logout timer. Set the “Log off idle users after” to 60 minutes 
Result and Comments	EdgeIPS initialized successful and new device IP address assigned.

Lab5: Register EdgeIPS with EdgeOne for centralized management

Table 9 Register EdgeIPS with EdgeOne for centralized management

Items	Descriptions
Application target	Host A
Steps	- Go to the EdgeIPS device web UI, go to Admin > System > System Time - Configure System Time Settings - If you are using the EdgeIPS in stand-alone mode, you need to setup NTP and the time zone here. In our lab setup, we will link the EdgeIPS to the EdgeOne server. Once this is configured, the EdgeIPS will sync its time from EdgeOne. - For this training, set the Time Zone to UTC 

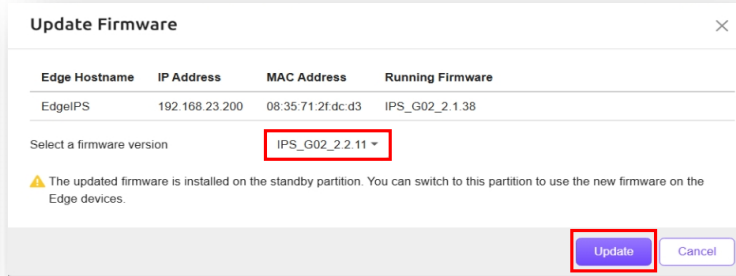
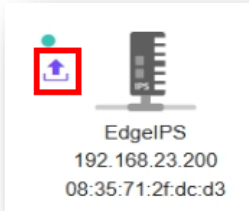
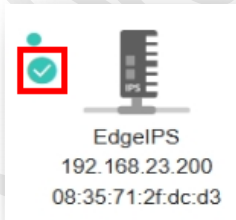
Items	Descriptions
	5. Go to Admin > System > EdgeOne Connection Settings and configure the EdgeOne Server Address: 192.168.23.21 and click Save.  6. The connection status will show “Connecting” and after a few seconds when it has successfully connected to EdgeOne, it will change to “Connected”. If it remains in “Connecting” then the EdgeIPS is not able to connect to EdgeOne. This can be caused by a firewall on your laptop or by Security Software that runs on your laptop. (try to disable it to allow incoming connections) If you can’t fix this, ask your instructor for help. 

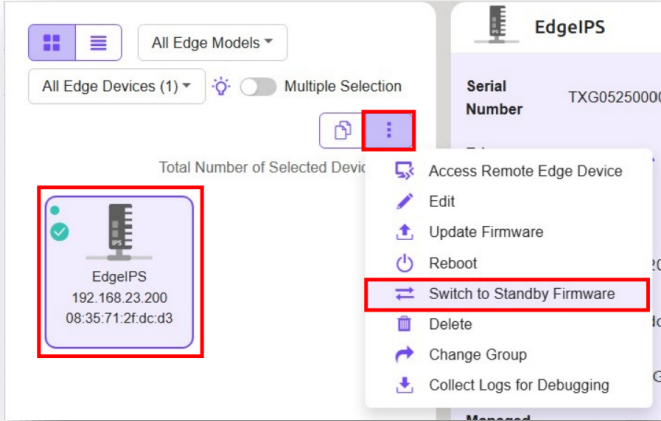
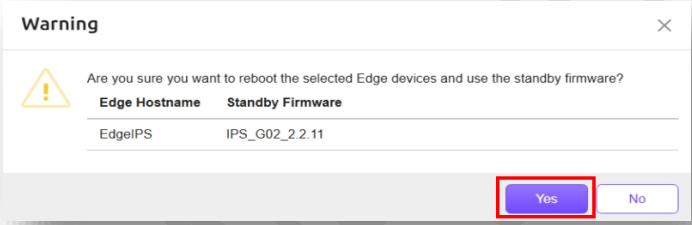
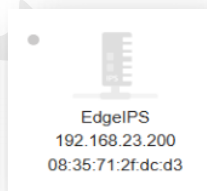
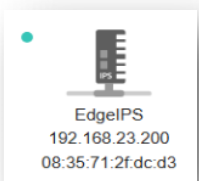
Items	Descriptions
	7. Check the EdgeIPS status LEDs, the “MGMT” LED should be green. This means the EdgeIPS is managed by EdgeOne and the connection between EdgeIPS and EdgeOne is alive. 
Expected Result	The EdgeIPS registration to EdgeOne successful

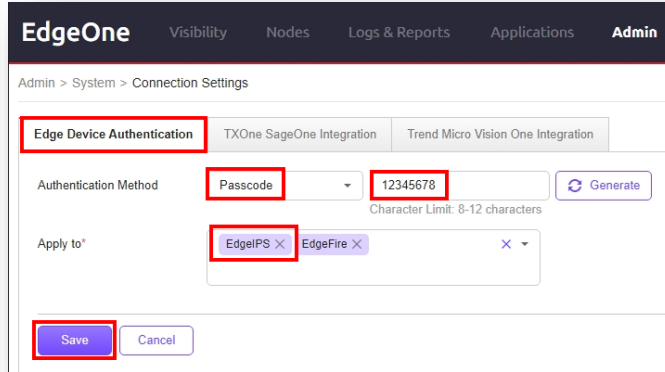
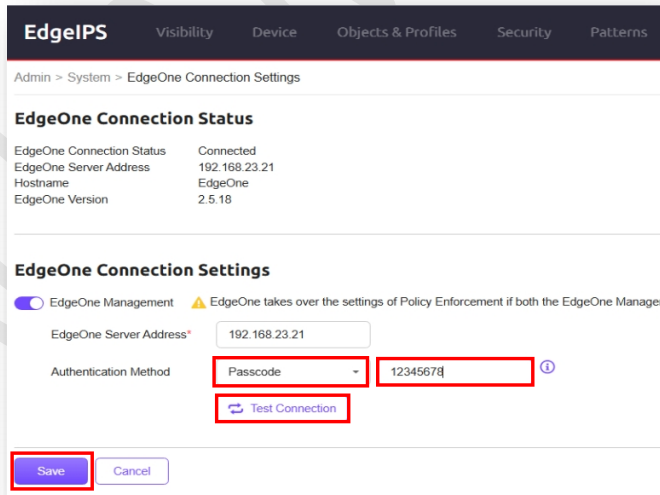
Lab6: Device Group Management

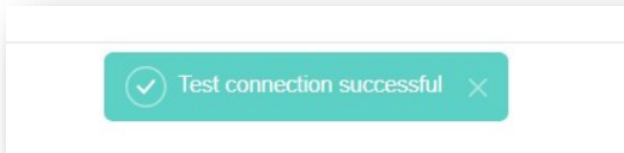
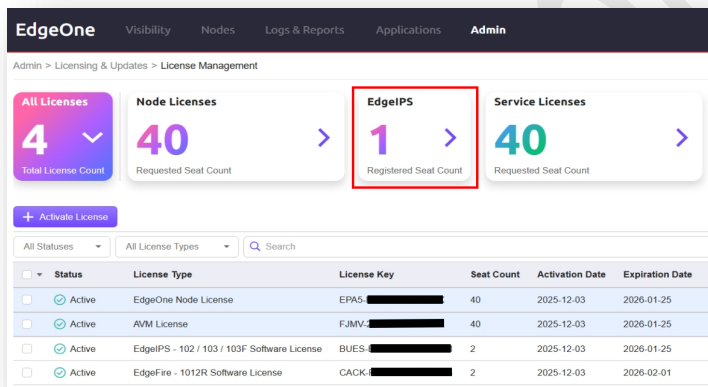
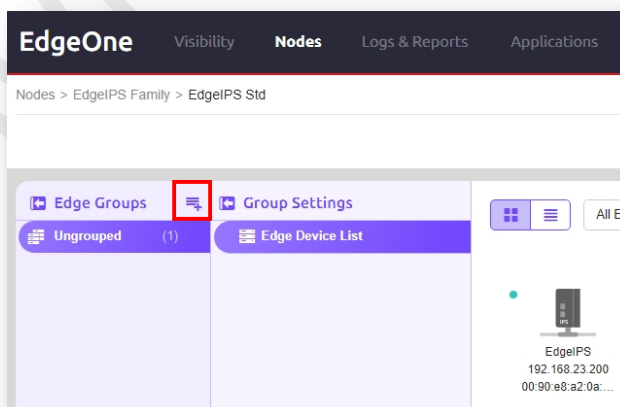
Table 10 Device Group Management

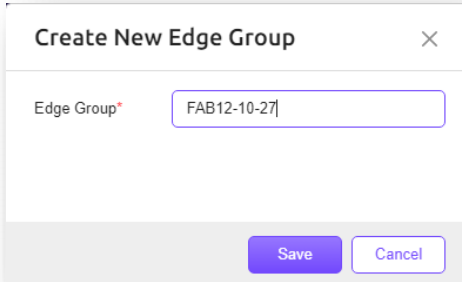
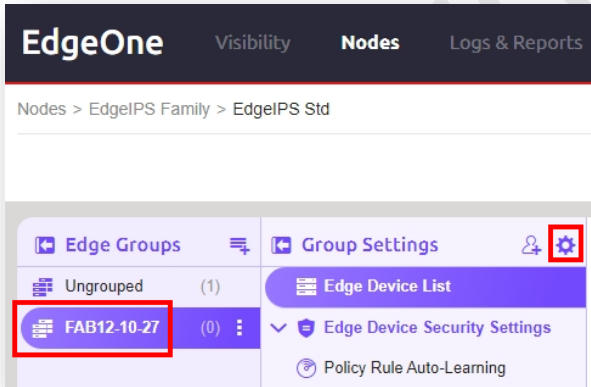
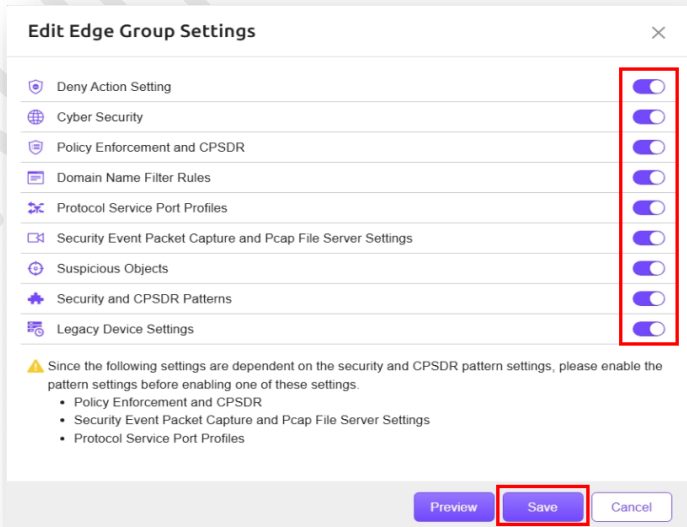
Items	Descriptions
Application target	Host A
Steps	- Log in to the VM LabConsole and navigate to https://192.168.23.21. Log in with the following credentials: User: txone Password: P@ssw0rd - Go to Node > EdgeIPS Family > EdgeIPS Std and check if a new EdgeIPS is listed under the Device Group "Ungrouped". Now upgrade the EdgeIPS to firmware version 2.2 or above. The screenshot shows the EdgeOne web interface. The breadcrumb navigation is 'Nodes > EdgeIPS Family > EdgeIPS Std'. Under the 'Edge Groups' tab, the 'Ungrouped' group is selected and highlighted with a red box, showing '(1)' device. To the right, the 'Edge Device List' shows a single device: 'EdgeIPS' with IP '192.168.23.200' and MAC '08:35:71:2f:dc:d3', also highlighted with a red box. - Click Upgrade firmware. The screenshot shows the 'All Edge Devices (1)' list. A device is selected and highlighted with a red box. A context menu is open over it, with the 'Update Firmware' option highlighted by a red box. Other options in the menu include 'Access Remote Edge Device', 'Edit', 'Reboot', 'Switch to Standby Firmware', 'Delete', 'Change Group', 'Copy Edge Device Policy Rules, Objects and Profiles', and 'Collect Logs for Debugging'.

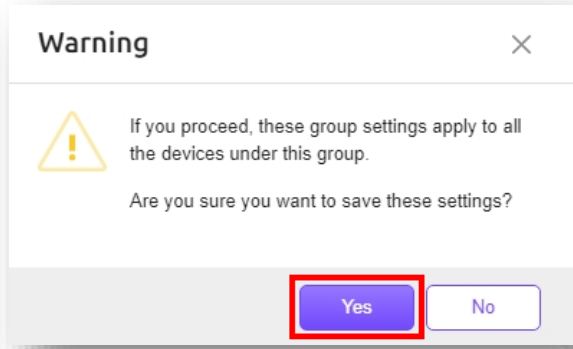
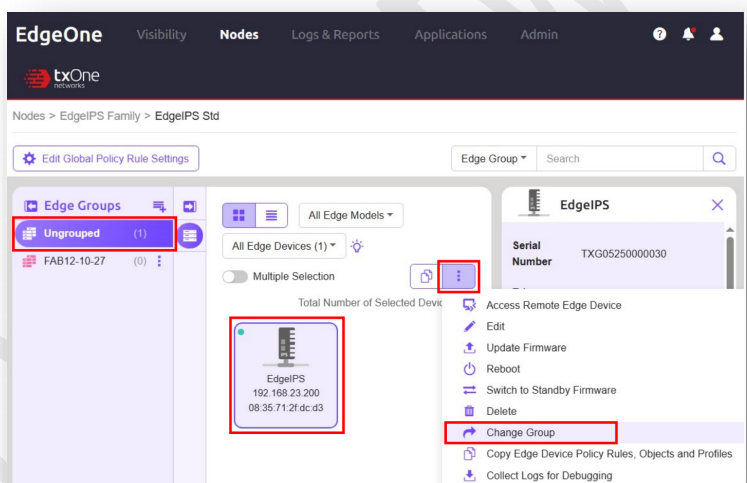
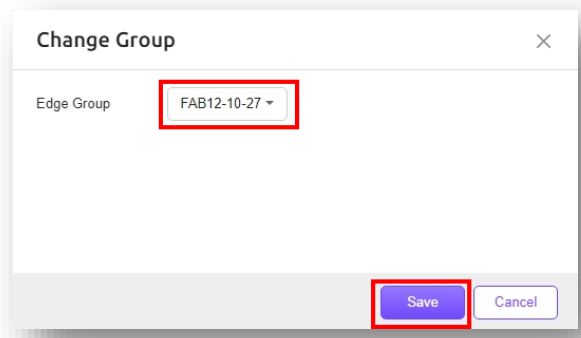
Items	Descriptions
	4. Click Update to upgrade the EdgeIPS 102 firmware to the latest version.  5. There will be a “Update Firmware” icon next to the EdgeIPS.  6. Once the upgrade task has completed, there will be a Green Check Mark.  7. Click Switch to Standby Firmware to switch the uploaded firmware from the “standby” partition to the “running” partition.

Items	Descriptions
	 8. Click Yes to switch it to the running partition.  9. The EdgeIPS device icon will become grayed-out during the firmware switching process.  10. The EdgeIPS will come back online once the firmware switch is complete. The warning badge will disappear, and a green indicator will appear on the device icon. 

Items	Descriptions
	11. Set the Edge Device Authentication Method to Passcode for a more secure authentication. Go to Admin > System > Connection Settings 12. Select the Edge Device Authentication Tab. Configure the authentication method to Passcode: 12345678, apply to EdgeIPS, EdgeFire and click Save.  13. Navigate to the EdgeIPS Device web UI https://192.168.23.200, go to Admin > System > EdgeOne Connection Settings. Configure the Authentication Method to the “Passcode”, Enter “12345678” and click Test Connection. 

Items	Descriptions
	14. The web UI will show the result of test. Click Save if the connection test is successful.  15. Navigate to the EdgeOne web UI https://192.168.23.21. Go to Admin > Licensing & Updates > License Management. You'll see there is 1 EdgeIPS seat count.  16. Go to Node > EdgeIPS Family > EdgeIPS Std and click “Create New Edge Group” . 

Items	Descriptions
	17. Name the group: "FAB12-10-27".  18. Select the Edge group "FAB12-10-27" and click Edit Edge Group Settings icon "⚙️".  19. Enable all features under Edit Group Settings and click Save. 

Items	Descriptions
	20. Click Yes to save the settings.  A warning dialog box with a yellow triangle icon. The text reads: "If you proceed, these group settings apply to all the devices under this group. Are you sure you want to save these settings?". At the bottom, there are two buttons: "Yes" (highlighted with a red box) and "No". Move the EdgeIPS from "Ungrouped" to "FAB12-10-27" 1. Select "Ungrouped" from the Edge Group list. 2. Select the EdgeIPS and click  Change Group.  A screenshot of the EdgeOne web interface. The "Edge Groups" list on the left shows "Ungrouped" (1) and "FAB12-10-27" (0). The "Ungrouped" group is selected. In the center, an EdgeIPS device is highlighted with a red box. On the right, a context menu is open, and the "Change Group" option is highlighted with a red box. 3. Move the EdgeIPS to the group "FAB12-10-27".  A "Change Group" dialog box. It has a label "Edge Group" and a dropdown menu showing "FAB12-10-27" (highlighted with a red box). At the bottom, there are two buttons: "Save" (highlighted with a red box) and "Cancel".